Northeastern University

Lecture 19: Project Information & Important Dates

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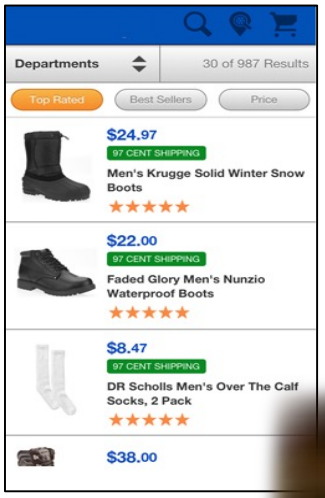
Goal of the Final Project

- Apply learned knowledge of data structures and algorithms to solve interesting problems



Requirements

- Each team has 3 students
- Required topics:

- Eclipse Java Project
- GUI (Graphical User Interface): Either using Java Swing or JavaFX.

Required topic in the list

- Bag (Chapter 1, 2, 3)
- The Efficiency of Algorithm (Chapter 4)
- Stacks (Chapter 5, 6)
- Recursion (Chapter 9, 14)

One topic from this list

- Queues, Deques, Priority Queue (Chapter 7, 8)
- Lists (Chapter 10, 11, 12)
- Sorting (Chapter 15, 16)
- Hashing (Chapter 22, 23)
- Trees (Chapter 24, 25)
- Binary Search Tree (Chapter 26)
- Heap (Chapter 27)
- Balanced Search Tree (Chapter 28)
- Graph (Chapter 29, 30)

(At least) Three topics from this list

Requirements (cont.)

- For the used data structure (such as Queue, Tree, Bag, etc.), the team needs to design the ADTs and implement ADTs as data structures. Then, using the implemented ADTs in the final project.

Important Dates

- Final Project Presentation (Online via Zoom)
  - » Period: Thursday April 9 - Tuesday April 14
  - » Final project presentation (PPT) + live demo (total: <= 30 mins)
  - » The presentation slides should cover following required sections :
    - Problem Description (Introduction)
    - Analysis (Your study, research, etc.)
    - Design (Algorithms, system design, flow chart, etc.)
    - Implementation (What APIs are used, what tools are used, etc.)
    - Results (Screenshots of sample runs, comparison, user study, etc.)
    - Discussion (Reflection)
    - Conclusions (and Future Work)
    - Job assignments (each member's contributions)

Important Dates (cont.)

- Final Project Submission
  - » Due: 11:59 pm, Tuesday, April 14
  - » 1. Term project source files (Eclipse project)
    - » Add an instruction file (.txt) to explain how to run the project.
  - » 2. Final project presentation slides (.pdf or .pptx file)
  - » All mentioned files should be in one folder and compressed as a .zip file. The file name should be "TeamName.zip".
  - » Submission link will be available on the Canvas later.